INCH-POUND

MIL-DTL-13791/22
15 January 2009

DETAIL SPECIFICATION SHEET

MACHINE-PAN HEAD SCREW, CROSS-RECESSED, CARBON STEEL, PROPER PROTECTED COATING PLATED, UNC-2A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-13791.

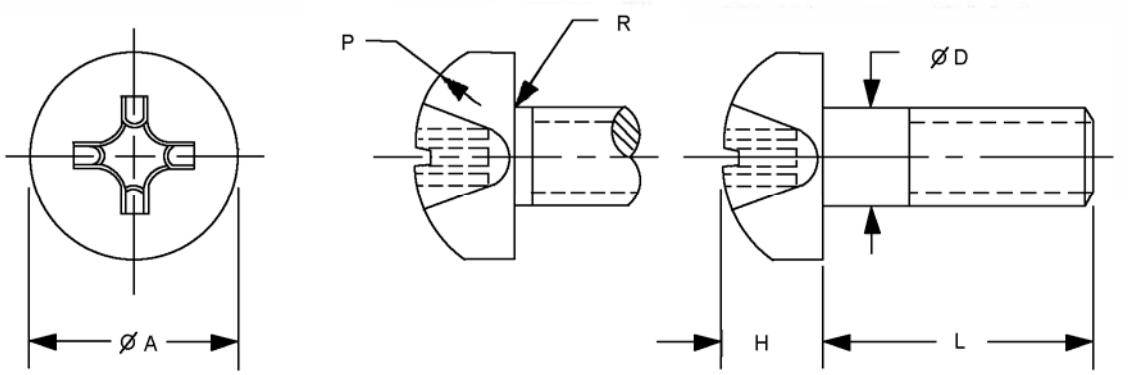


FIGURE 1. Machine-pan head screw, cross-recessed, carbon steel, with proper protected coating type plated, UNC-2A.

AMSC N/A

FSC 2920

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TABLE I. Dash numbers and dimensions. 1/ 2/

SIZE		.086 (2.18)	.112 (2.84)	.138 (3.51)	.164 (4.17)	.190 (4.83)	.250 (6.35)	.3125 (7.94)	.375 (9.53)
THREADS PER INCH UNC-2A		56	40	32	32	24	20	18	16
D BODY	MAX	.0860 (2.18)	.1120 (2.84)	.1380 (3.51)	.1640 (4.17)	.1900 (4.83)	.2500 (6.35)	.3125 (7.94)	.3750 (9.53)
	MIN	.0717 (1.82)	.0925 (2.35)	.1141 (2.90)	.1399 (3.55)	.1586 (4.03)	.2127 (5.40)	.2712 (6.89)	.3287 (8.35)
A HEAD	MAX	.1670 (4.24)	.2190 (5.56)	.2700 (6.86)	.3220 (8.18)	.3730 (9.47)	.4920 (12.50)	.6150 (15.62)	.7400 (18.80)
	MIN	.1550 (3.94)	.2050 (5.21)	.2560 (6.50)	.3060 (7.77)	.3570 (9.07)	.4730 (12.01)	.5940 (15.09)	.7160 (18.19)
H HEAD HEIGHT	MAX	.0620 (1.57)	.0800 (2.03)	.0970 (2.46)	.1150 (2.92)	.1330 (3.38)	.1750 (4.45)	.2180 (5.54)	.2610 (6.63)
	MIN	.0530 (1.35)	.0700 (1.78)	.0870 (2.21)	.1050 (2.67)	.1220 (3.10)	.1620 (4.11)	.2030 (5.16)	.2440 (6.20)
P HEAD RADIUS	MIN	.0100 (.25)	.0100 (.25)	.0150 (.38)	.0150 (.38)	.0200 (.51)	.0350 (.89)	.0400 (1.02)	.0400 (1.02)
R FILLET RADIUS	MAX	.0130 (.33)	.0170 (.43)	.0210 (.53)	.0250 (.64)	.0290 (.74)	.0380 (.97)	.0470 (1.19)	.0560 (1.42)
TENSILE STRENGTH LUS	MIN	220	360	545	840	1,050	1,900	3,140	4,65
L LENGTH	TOLERANCE	DASH NUMBER	DASH NUMBER	DASH NUMBER	DASH NUMBER	DASH NUMBR	DASH NUMBER	DASH NUMBR	DASH NUMBER
.125 (3.18) .188 (4.78) .250 (6.35) .312 (7.92) .375 (9.53) .438 (11.13) .500 (12.70) .562 (14.27) .625 (15.88) .750 (19.05) .875 (22.23) 1.000	+.000 (.00) -.031 (.79)	201 202 203 204 205 206 207 320 3/ 208 209 210 321 3/ 322 3/	211 212 213 214 215 216 217 323 3/ 218 219 220 221 324 3/ 222 325 3/ 223 326 3/	224 225 226 227 228 229 230 327 3/ 231 232 233 234 328 235 329 3/ 236 237 361 3/ 238 330 3/ 362	239 240 241 242 243 244 245 331 3/ 246 247 248 249 332 250 333 3/ 251 252 334 3/ 253 254 255 255 257 335 3/ 336 3/	258 259 260 261 262 263 337 3/ 264 265 266 267 338 268 339 3/ 269 270 340 3/ 271 272 273 274 275 341 3/ 342 3/	276 277 278 279 343 3/ 280 281 282 283 344 284 245 3/ 285 286 246 3/ 287 289 289 290 291 347 3/ 348 3/	292 293 294 349 3/ 295 296 297 298 350 299 351 3/ 300 301 352 3/ 302 303 304 305 306 353 3/ 354 3/	307 355 3/ 308 309 310 311 356 312 357 3/ 313 314 358 3/ 315 316 317 318 319 359 3/ 360 3/
1.125 (28.58) 1.250 (31.75) 1.375 (34.93) 1.500 (38.10) 1.750 (44.45) 1.875 (47.63) 2.000 (50.80)	+.000 (.00) -.062 (1.57)								
2.250 (57.15) 2.500 (63.50) 2.750 (69.85) 3.000 (76.20) 3.250 (82.55) 3.500 (88.90)	+.000 (.00) -.094 (2.39)								

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ Indicates manufacturer's non-stock production items.

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REQUIREMENTS

Cadmium is not recommended. It is recommended that the use of carbon steel material with cadmium plating is used only when the other materials and plating finishes specified cannot meet performance requirements. (See table ll, NASM35206 and MIL-DTL-13791) It should be a proper protected coating type, and there are many alternative materials, which are Zinc, Zinc/Nickel, Zinc/Cobalt, Tin/Zinc, for cadmium plating applications. Proper protected coating type plating shall be in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II class 2. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Dash numbers: See table l.

Dimensions and configurations: See figure 1.

Part or Identifying Number (PIN) example: M13791/22-Q.

See table ll for dash number code.

TABLE ll. Material and finish. 1/

Dash number	Material	Finish
Q	Carbon steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class. 2/
Z	Carbon steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 25, or ASTM B695, type II, class 5. 3/
S	300 series stainless steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.

1/ Must be capable of withstanding 96 hours salt spray.
2/ Not recommended for use where severe abrasion conditions exist. Max service temperature 120°C.
3/ Embitterment test need not be run.

Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein.

Referenced documents. In addition to MIL-DTL-13791, this document references the following:

ASTM B633
ASTM B695
NASM35206
SAE-AMS2700
SAE-AMS-C-81562
SAE-AMS-QQ-P-416

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 2920-2008-022)

Review activities:
Army - CR, CR4
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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